Q1. DIRECTIONS *for question 1 to 3:* Type in your answer in the input box provided below the question.

If ABCD is a square of side 24 cm and an isosceles triangle PCD is drawn, with PC = PD, such that the area common to the square ABCD and the triangle PCD is 360 sq.cm, find the area (in sq.cm) of the triangle PCD.

Q2. DIRECTIONS *for question 1 to 3:* Type in your answer in the input box provided below the question.

N is a natural number greater than 1. A and B are single-digit natural numbers, with $A \geq B$, such that for any value of N, $(A + B)^N$ has the same units digit as $A + B$ and $(A \times B)^N$ has the same units digit as $(A \times B)$. How many pairs of values of A and B exist satisfying these conditions?

Q4. DIRECTIONS *for question 4:* Select the correct alternative from the given choices.

Find the equation of the line parallel to the line $2x + 3y = 6$ and having a *y* intercept of 5.

- a) $2x - 3y = 11$
- b) $2x + 3y = 10$
- c) **$2x + 3y = 15$**
- d) $2x + 3y = 5$

Q5. DIRECTIONS *for question 5:* Type in your answer in the input box provided below the question.

How many three-digit numbers exist such that the ratio of the hundreds digit and the tens digit is the same as the ratio of the tens digit and the units digit?

Q6. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.
If 15/08/1997 was a Friday, 15/08/1947 must have been a

- a) Tuesday
- b) Wednesday
- c) **Thursday**
- d) Friday

Q7. **DIRECTIONS** *for questions 6 and 7:* Select the correct alternative from the given choices.

If the 11^{th} and the 8^{th} terms of an arithmetic progression are in the ratio 2 : 5, which of the following statements is definitely false?

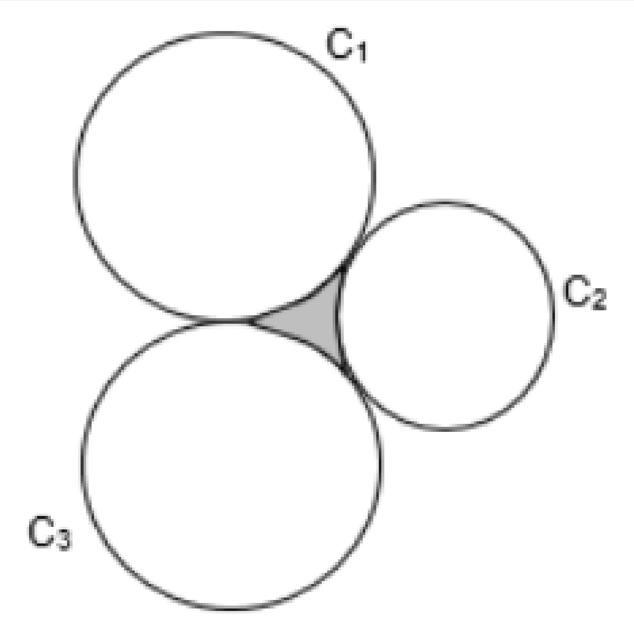
- a) The sum of no two consecutive terms is zero.
- b) The sum of no three consecutive terms is zero.
- c) **At least one of the terms is zero.**
- d) None of the above

Q8. DIRECTIONS *for question 8:* Type in your answer in the input box provided below the question.

Anju and Bobby simultaneously started climbing up an ascending escalator (a moving staircase). Since they were in a hurry, they also started climbing up the steps (taking one step at a time). Anju took k steps in the time that Bobby took 1 step, where k is an integer greater than 1. How many values can k assume, if it is known that the speed of the escalator was the same as that of Bobby and Anju took a total of 72 steps to climb up the escalator? (Assume that the number of steps taken by Bobby was also an integer)

Q9. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.

Find the area (in sq. cm) of the shaded region in the figure below, if the circles C_1, C_2 and C_3 touch each other externally as shown and have radii measuring 9 cm, $9(\sqrt{2}-1)$ cm and 9 cm respectively.



a)

$\frac{81}{4}(3\pi - 2\sqrt{2})$

b) $81\left(1 - \pi + \frac{\pi}{\sqrt{2}}\right)$

c) $\frac{81}{2}(2\pi - \sqrt{2})$

d)

$81(1 - 2\pi + \sqrt{3}\pi)$

Q10. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.

There are two drums D_1 and D_2, each of which is filled to the brim with water. Now, a leak is made at the bottom of each of D_1 and D_2, such that the leak in D_1 takes 6 hours to empty it, while the leak in D_2 takes 9 hours to empty it. If the capacity of D_1 is more than the capacity of D_2 by 60%, then find the time after which the volume of water in D_2 will be 25% more than the volume of water in D_1.

- a) $2\frac{17}{23}$ hours
- b) $4\frac{1}{2}$ hours
- c) **3 hours**
- d) $4\frac{2}{3}$ hours

Q11. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.

If the product and the sum of all possible values of *x* satisfying the equation $3x^3 - 31x^2 + 76x - 22 = 0$ are denoted by P and S respectively, find the value of S – P.

- a) 3
- b) $\frac{4}{3}$
- c) **5**
- d) $\frac{16}{3}$

Q12. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.

2*n* years ago, the age of Raju was four times that of his son and *n* years ago, the age of Raju was thrice that of his son. If *n* years later, the sum of the ages of Raju and his son will be 80 years, then the difference in the ages of Raju and his son is

- a) 20 years.
- b) 40 years.
- c) **24 years.**
- d) 30 years.

Q13. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.

A normal pack of 52 cards comprises four suits – Spades, Clubs, Diamonds and Hearts − of 13 cards each, the Spades and Clubs being black and the other two red. Further, each suit of 13 cards comprises nine numbered cards − numbered from 2 to 10 − and four honour cards − Ace, King, Queen and Jack. Now, it is known that a card is missing from a normal pack of 52 cards. If from that pack, two black cards can be selected in 300 ways and two numbered cards can be selected in 630 ways, which of the following could be the missing card?

- a) Spade Ace
- b) Diamond king
- c) **Club king**
- d) More than one of the above

Q14. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.
Pranav writes in his notebook a number, N, to the base 16. He observes that N is a nine-digit number and the sum of these nine digits is $(30)_{10}$. If sum of the digits in the even places is equal to the sum of the digits in the odd places, then $(N)_{16}$ is always divisible by

a) $(99)_{10}$

b) $(99)_{16}$

c) **$(255)_{10}$**

d) $(255)_{16}$

Q15. **DIRECTIONS** *for questions 9 and 16:* Select the correct alternative from the given choices.

A certain sum lent under simple interest doubles itself in five years. After how many years will the sum become four times itself, if the rate of interest at which it is lent is increased by 5 percentage points?

- a) 12
- b) 16
- c) **20**
- d) 15

Q16. DIRECTIONS *for questions 9 and 16:* Select the correct alternative from the given choices.

Ramu had two jars, one of which had 1 litre of 40% milk and the other, 1 litre of 18% milk. If he mixed 300 ml of the former and 700 ml of the latter to form a new solution, what is the ratio of water and milk in the solution?

a) 32 : 93

b) 377 : 123

c) **417 : 83**

d) 747 : 253

Q17. DIRECTIONS *for questions 17 and 18:* Type in your answer in the input box provided below the question.

Exactly 3000 students wrote a multiple-choice test comprising exactly five questions, each question, in turn, having exactly five answer choices, exactly one of which is correct. Every question correctly answered fetches 4 marks and every question incorrectly answered fetches –1 mark. If each student attempted all the questions and the answer choices marked by no two students are the same for all the five questions, then the number of students with a net positive score in the test is at least

Q18. **DIRECTIONS** *for questions 17 and 18:* Type in your answer in the input box provided below the question.

If $51 + 52 + 53 + \dots + N = 5985$, then find the value of $N^2 + N$.

Q19. DIRECTIONS *for question 19:* Select the correct alternative from the given choices.

Let S be a series of natural numbers containing n terms. S_o is the sum of every alternate term, starting from the first term and S_e is the sum of all the remaining terms. If the product of S_o and S_e is odd, then which of the following statements is necessarily true?

I. n is odd.

II. n is even.

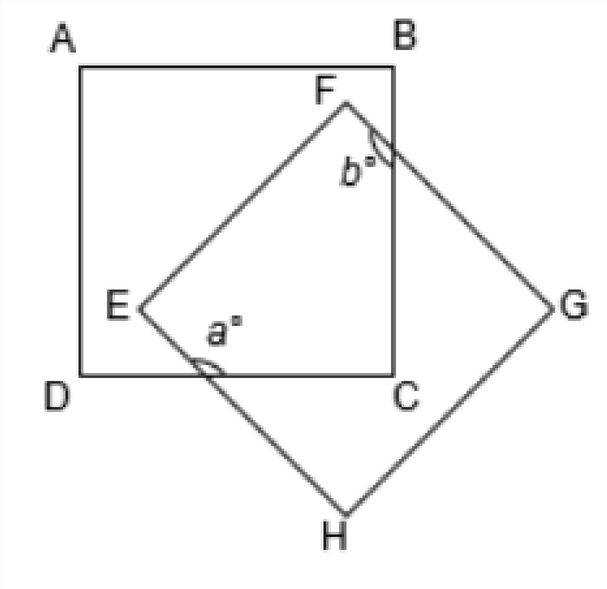
III. The difference in the number of terms in S that are odd and the number of terms in S that are even is odd.

IV. The number of terms in S that are odd is even.

- a) Only I and III
- b) Only III
- c) **Only II and IV**
- d) Only IV

Q20. DIRECTIONS *for question 20:* Type in your answer in the input box provided below the question.

In the figure below, if ABCD and EFGH are two overlapping squares, find the value of (a + b) (in degrees).



Q3. DIRECTIONS *for question 1 to 3:* Type in your answer in the input box provided below the question.

If x is an integer and $x^3 - 2(x + 2)^2 = 993$, find the value of x.

Q21. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

A milkman purchases milk for ₹18 per litre but adds one litre of water to every 1.5 litres of milk that he purchases and then sells the mixture to his customers at the same price per litre. What is the percentage of profit that he makes in selling the mixture?

- a) 40%
- b) 50%
- c) **66.67%**
- d) 33.33%

Q22. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

If the inequality $\dfrac{x^2 - \frac{2x}{3} + 1}{ax^2 + bx + 1} \geq 0$ is true if an only if $x < -2$ or $x > \frac{-1}{2}$, find $(a + b)$.

- a) 4
- b) 4.5
- c) **3.5**
- d) 3